The Chinese University of Hong Kong (Shenzhen)
ADVANCED MATH FOR AI
Fall 2026

Assignment 2

Issued: Sunday 27th September, 2026 **Due:** Sunday 18th October, 2026

Total: 100 points, plus 10 bonus points.

1. Matrix Diagonalizability **(25 points)**

An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.

(a) For each matrix below, find the eigenvalues and a basis for each eigenspace by solving $(A - \lambda I)x = 0$. Then determine whether the matrix is diagonalizable and justify your answer.

Hint: The first matrix is block upper triangular. The determinant of a block triangular matrix is the product of the determinants of its diagonal blocks.

i. (10 points)

$$A_1 = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

ii. (10 points)

$$A_2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(b) (5 points) Neither diagonalizability nor invertibility implies the other. Determine whether each of the two matrices above is invertible, and briefly explain how the two examples illustrate this statement.

Solution:

(a) i. Since A_1 is block upper triangular,

$$\det(A_1 - \lambda I) = (1-\lambda)\det\begin{bmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{bmatrix} = (1-\lambda)(\lambda-1)(\lambda-5).$$

Thus the eigenvalues are $\lambda = 1$, with algebraic multiplicity 2, and $\lambda = 5$, with algebraic multiplicity 1.

For $\lambda = 1$,

$$A_1 - I = \begin{bmatrix} 1 & 3 & 0 \\ 1 & 3 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

The equations give $x_1 = -3x_2$ and $x_3 = 0$, so

$$E_1 = \text{Null}(A_1 - I) = \text{span}\left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Hence $\dim(E_1) = 1$, which is smaller than the algebraic multiplicity 2.
For $\lambda = 5$,

$$A_1 - 5I = \begin{bmatrix} -3 & 3 & 0 \\ 1 & -1 & 3 \\ 0 & 0 & -4 \end{bmatrix},$$

so $x_3 = 0$ and $x_1 = x_2$. Therefore

$$E_5 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

In total, A_1 has only $1+1 = 2$ linearly independent eigenvectors. Therefore A_1 is not diagonalizable.

ii. Since A_2 is upper triangular, its eigenvalues are its diagonal entries: $\lambda = 1$, with algebraic multiplicity 1, and $\lambda = 0$, with algebraic multiplicity 3.

For $\lambda = 1$, solving $(A_2 - I)x = 0$ gives $x_2 = x_3 = x_4 = 0$, with x_1 free. Hence

$$E_1 = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda = 0$, the equation $A_2 x = 0$ is $x_1 + x_2 = 0$. Thus

$$E_0 = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

The two eigenspaces provide $1+3 = 4$ linearly independent eigenvectors, so A_2 is diagonalizable.

(b) We have

$$\det(A_1) = 1 \cdot \det \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} = 8 - 3 = 5 \neq 0,$$

so A_1 is invertible. Since A_2 has three zero rows, $\det(A_2) = 0$, so A_2 is not invertible. Thus A_1 is an invertible matrix that is not diagonalizable, while A_2 is a diagonalizable matrix that is not invertible. These examples show that neither property implies the other.

2. Matrix Norms **(35 points)**

This problem introduces matrix norms, a useful set of tools for measuring matrix size and approximation error in machine learning. We will also derive the Cauchy–Schwarz inequality for Euclidean vectors and establish useful bounds involving the

spectral and Frobenius norms.

Let $x, y \in \mathbb{R}^n$, with $\langle x, y\rangle = x^\mathsf{T} y$ and $\|x\|_2 = \sqrt{\langle x, x\rangle}$.

(a) (10 points) For $A \in \mathbb{R}^{m\times n}$, define its spectral norm by

$$\|A\|_2 = \max_{z\neq 0} \frac{\|Az\|_2}{\|z\|_2}.$$

Show that $\|A\|_2 = \sigma_1$, the largest singular value of A.

(b) (10 points) Define the Frobenius norm by

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2} = \sqrt{\mathrm{tr}(A^\mathsf{T}A)}.$$

If $r = \mathrm{rank}(A)$, show that

$$\|A\|_F^2 = \sum_{i=1}^{r} \sigma_i^2.$$

(c) (8 points) Prove the Cauchy–Schwarz inequality

$$|\langle x, y\rangle| \leq \|x\|_2 \|y\|_2.$$

Hint: When $y \neq 0$, use $\|x - ty\|_2^2 \geq 0$ for every $t \in \mathbb{R}$, and choose a suitable value of t.

(d) (7 points) Prove that, for every $x \in \mathbb{R}^n$,

$$\|Ax\|_2 \leq \|A\|_2 \|x\|_2 \leq \|A\|_F \|x\|_2.$$

Clearly indicate where you use the definition of the spectral norm and where you use the relationship between the spectral and Frobenius norms.

Solution: Let

$$A = U\Sigma V^\mathsf{T}$$

be a singular value decomposition, where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$.

(a) For any nonzero z, set $w = V^\mathsf{T} z$. Since V is orthogonal, $\|w\|_2 = \|z\|_2$, and

$$\|Az\|_2^2 = \|U\Sigma w\|_2^2 = \|\Sigma w\|_2^2 = \sum_i \sigma_i^2 w_i^2 \leq \sigma_1^2 \sum_i w_i^2 = \sigma_1^2 \|z\|_2^2.$$

Therefore $\|Az\|_2/\|z\|_2 \leq \sigma_1$. Equality is attained by taking $z = v_1$, because $\|v_1\|_2 = 1$ and $Av_1 = \sigma_1 u_1$. Hence

$$\|A\|_2 = \sigma_1.$$

(b) Recall the cyclic property of the trace: whenever the matrix products are defined,

$$\operatorname{tr}(X_1 X_2 \cdots X_p) = \operatorname{tr}(X_2 \cdots X_p X_1).$$

Thus, factors inside a trace may be moved cyclically from one end to the other (but they may not be rearranged arbitrarily). Using this property and the orthogonality of U and V,

$$\begin{aligned}\|A\|_F^2 &= \operatorname{tr}(A^\mathsf{T} A) \\ &= \operatorname{tr}\big(V\Sigma^\mathsf{T} U^\mathsf{T} U \Sigma V^\mathsf{T}\big) \\ &= \operatorname{tr}\big(V\Sigma^\mathsf{T}\Sigma V^\mathsf{T}\big) \\ &= \operatorname{tr}\big(\Sigma^\mathsf{T}\Sigma V^\mathsf{T} V\big) = \operatorname{tr}(\Sigma^\mathsf{T}\Sigma) = \sum_{i=1}^{r} \sigma_i^2.\end{aligned}$$

(c) If $y = 0$, both sides are zero, so the result is immediate. Suppose $y \neq 0$. For every $t \in \mathbb{R}$,

$$0 \leq \|x - ty\|_2^2 = \|x\|_2^2 - 2t\langle x, y\rangle + t^2\|y\|_2^2.$$

Choose $t = \langle x, y\rangle / \|y\|_2^2$. Substitution gives

$$0 \leq \|x\|_2^2 - \frac{\langle x, y\rangle^2}{\|y\|_2^2}.$$

Thus

$$\langle x, y\rangle^2 \leq \|x\|_2^2 \|y\|_2^2.$$

Taking square roots yields

$$|\langle x, y\rangle| \leq \|x\|_2 \|y\|_2.$$

(d) If $x = 0$, the inequalities are immediate. If $x \neq 0$, the definition of the spectral norm gives

$$\frac{\|Ax\|_2}{\|x\|_2} \leq \|A\|_2, \qquad \text{so} \qquad \|Ax\|_2 \leq \|A\|_2 \|x\|_2.$$

Parts (a) and (b) imply

$$\|A\|_2 = \sigma_1 \leq \sqrt{\sum_{i=1}^{r} \sigma_i^2} = \|A\|_F.$$

Multiplying by $\|x\|_2 \geq 0$ gives

$$\|A\|_2 \|x\|_2 \leq \|A\|_F \|x\|_2.$$

Combining the two inequalities proves the claim.

3. Best Rank-k Approximation (40 + 10 bonus points)

In class, we introduced the truncated SVD as a way to approximate a matrix using a lower-rank matrix. This problem explains why it is optimal: no other matrix of rank at most k can achieve a smaller error.

Let $A \in \mathbb{R}^{m\times n}$ have rank r and singular value decomposition

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^{\mathsf{T}}, \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0.$$

For $k < r$, the rank-k truncated SVD is

$$\widehat{A}_k = \sum_{i=1}^{k} \sigma_i u_i v_i^{\mathsf{T}}.$$

(a) (15 points) For

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix},$$

find a singular value decomposition and use it to compute the best rank-1 approximation of A.

(b) (10 points) Show that

$$\|A - \widehat{A}_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2.$$

(c) (8 points) Directly from the definition of the spectral norm, show that

$$\|A - \widehat{A}_k\|_2 = \sigma_{k+1}.$$

(d) (10 points (bonus)) Let $B \in \mathbb{R}^{m\times n}$ satisfy $\operatorname{rank}(B) \leq k$. Prove that

$$\|A - B\|_2 \geq \sigma_{k+1}.$$

Hint: Let $V_{k+1} = [v_1, \ldots, v_{k+1}]$. Use the rank–nullity theorem to find a nonzero vector in $\operatorname{Null}(B) \cap \operatorname{Col}(V_{k+1})$.

(e) (7 points) Using part (c) and the inequality stated in part (d), conclude that

$$\widehat{A}_k \in \operatorname*{arg\,min}_{\operatorname{rank}(B)\leq k} \|A - B\|_2.$$

You may use the inequality in part (d) even if you did not complete the bonus proof. Briefly explain what this result says about the truncated SVD.

Solution:

(a) Following the SVD construction, first compute

$$A^{\mathsf{T}}A = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}.$$

Its eigenvalues are 25, 9, and 0. Corresponding orthonormal eigenvectors are

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \frac{1}{3\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}, \qquad v_3 = \frac{1}{3} \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}.$$

Therefore the nonzero singular values are $\sigma_1 = \sqrt{25} = 5$ and $\sigma_2 = \sqrt{9} = 3$. Using $u_i = Av_i/\sigma_i$ gives

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Thus one full SVD is

$$A = U\Sigma V^{\mathsf{T}},$$

where

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix},$$

and

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{3\sqrt{2}} & -\frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} & \frac{2}{3} \\ 0 & \frac{4}{3\sqrt{2}} & \frac{1}{3} \end{bmatrix}.$$

The best rank-1 approximation is

$$\widehat{A}_1 = \sigma_1 u_1 v_1^{\mathsf{T}} = \frac{5}{2} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

(b) Let Σ_k be obtained from Σ by keeping only its first k singular values, and put $D = \Sigma - \Sigma_k$. Then

$$A - \widehat{A}_k = U\Sigma V^{\mathsf{T}} - U\Sigma_k V^{\mathsf{T}} = UDV^{\mathsf{T}}.$$

The diagonal entries of D are $0, \ldots, 0, \sigma_{k+1}, \ldots, \sigma_r$, with any remaining entries equal to zero. Starting from the definition $\|M\|_F^2 = \operatorname{tr}(M^{\mathsf{T}}M)$, we obtain

$$\begin{aligned} \|A - \widehat{A}_k\|_F^2 &= \operatorname{tr}\big((UDV^{\mathsf{T}})^{\mathsf{T}} UDV^{\mathsf{T}}\big) \\ &= \operatorname{tr}\big(VD^{\mathsf{T}}U^{\mathsf{T}}UDV^{\mathsf{T}}\big) \\ &= \operatorname{tr}\big(VD^{\mathsf{T}}DV^{\mathsf{T}}\big) \qquad (U^{\mathsf{T}}U = I) \\ &= \operatorname{tr}\big(D^{\mathsf{T}}DV^{\mathsf{T}}V\big) \qquad \text{(cyclic property of trace)} \\ &= \operatorname{tr}(D^{\mathsf{T}}D) \qquad (V^{\mathsf{T}}V = I) \\ &= \sum_{i=k+1}^{r} \sigma_i^2. \end{aligned}$$

The last equality follows because $D^\top D$ is diagonal, with nonzero diagonal entries $\sigma_{k+1}^2, \ldots, \sigma_r^2$.

(c) Let $R = A - \widehat{A}_k$. Extend $\{v_1, \ldots, v_r\}$ to an orthonormal basis of $\mathbb{R}^n$, and write a unit vector x as $x = \sum_i \alpha_i v_i$, where $\sum_i \alpha_i^2 = 1$. Then

$$Rx = \sum_{i=k+1}^{r} \sigma_i \alpha_i u_i,$$

so

$$\|Rx\|_2^2 = \sum_{i=k+1}^{r} \sigma_i^2 \alpha_i^2 \leq \sigma_{k+1}^2 \sum_{i=k+1}^{r} \alpha_i^2 \leq \sigma_{k+1}^2.$$

Hence $\|R\|_2 \leq \sigma_{k+1}$. Taking $x = v_{k+1}$ gives

$$\|Rv_{k+1}\|_2 = \|\sigma_{k+1} u_{k+1}\|_2 = \sigma_{k+1},$$

so the upper bound is attained and

$$\|A - \widehat{A}_k\|_2 = \sigma_{k+1}.$$

(d) Let

$$S = \operatorname{span}\{v_1, \ldots, v_{k+1}\}.$$

Since $\operatorname{rank}(B) \leq k$, the rank–nullity theorem gives

$$\dim \operatorname{Null}(B) \geq n - k.$$

Because $\dim S = k + 1$,

$$\dim \operatorname{Null}(B) + \dim S \geq n + 1 > n.$$

Thus $\operatorname{Null}(B) \cap S$ contains a nonzero vector. Choose a unit vector x in this intersection. Then $Bx = 0$, and $x = \sum_{i=1}^{k+1} \alpha_i v_i$, with $\sum_{i=1}^{k+1} \alpha_i^2 = 1$. Therefore

$$\begin{aligned} \|(A - B)x\|_2^2 &= \|Ax\|_2^2 \\ &= \sum_{i=1}^{k+1} \sigma_i^2 \alpha_i^2 \\ &\geq \sigma_{k+1}^2 \sum_{i=1}^{k+1} \alpha_i^2 = \sigma_{k+1}^2. \end{aligned}$$

It follows that

$$\|A - B\|_2 \geq \|(A - B)x\|_2 \geq \sigma_{k+1}.$$

(e) Part (c) shows that the truncated SVD satisfies

$$\|A - \widehat{A}_k\|_2 = \sigma_{k+1}.$$

Part (d) shows that every matrix B of rank at most k satisfies

$$\|A - B\|_2 \geq \sigma_{k+1}.$$

Therefore $\widehat{A}_k$ attains the smallest possible spectral-norm error:

$$\widehat{A}_k \in \underset{\text{rank}(B)\leq k}{\arg\min} \|A - B\|_2.$$

In words, among all matrices of rank at most k, the truncated SVD is a best approximation to A in the spectral norm.

Remark. The analogous optimality result also holds for the Frobenius norm.